

Total no. of Pages:

Roll no.

Degree: B.Tech. Semester: III
MID-SEMESTER EXAMINATION, September 2025

Course Title: Probability Theory and Random Process
Course Code: ECECC303/EIECC303/ECECC06/EIECC06

Duration: 1:30 Hours

Max. Marks: 20

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Simplify: i) $\overline{A+B} + \overline{A+B}$ ii) $(A+B)(AB)$	1	1
1b	A ternary communication system is shown above. Suppose that input symbols 0, 1, and 2 occur with probability 1/3 respectively. (i) Find the probabilities of the output symbols. (ii) Suppose that a 1 is observed at the output. What is the probability that the input was 0?	3	1
2a	A die is tossed and the random variable X is defined as the number of full pairs of dots in the face showing up. i) Explain the range of X. ii) Find the PMF and CDF of X.	1	1
2b	A discrete random variable X is uniformly distributed in the set $\{-3, \dots, 3, 4\}$ i) Find the mean and variance of X. ii) Find the mean and variance of $Y = -2X^2 + 3$. iii) Find the mean and variance of $W = \cos(\pi X/8)$.	3	2
3a	Find the moment generating function of a Poisson random variable X. Further using the MGF, find the mean of X.	1	2
3b	A binary transmission system sends a "0" bit by transmitting a -V signal, and a "1" bit by transmitting a +V. The received signal is corrupted by noise and given by $Y=X+N$, where X is the transmitted signal, and N is a Gaussian noise with mean 0 and variance $N_0/2$. Assuming the probability of transmission of 1 is "p," find the PDF of Y.	3	3

4a	Find the PDF of a bivariate random variable (X, Y) , where X and Y are independently and identically distributed Gaussian random variables.	1	3												
4b	The joint PDF of a bivariate random variable (X, Y) is given by $f_{X,Y}(x,y) = \begin{cases} \frac{1}{y} e^{-\frac{x}{y}} e^{-y} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$ i) Show that $f_{X,Y}(x,y)$ is a valid joint PDF? ii) Find $P(X > 1 Y = y)$.	3	3												
5a	A university records the number of hours studied (X) and marks obtained (Y) by the students: <table><tr><td>X (hours)</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>Y (marks)</td><td>11</td><td>19</td><td>25</td><td>30</td><td>40</td></tr></table> Prove that X and Y share a very strong positive linear relationship.	X (hours)	2	4	6	8	10	Y (marks)	11	19	25	30	40	1	2
X (hours)	2	4	6	8	10										
Y (marks)	11	19	25	30	40										
5b	Show that if $Y = aX + b$, then prove that the correlation coefficient $\rho_{XY} = \text{sgn}(a)$, when $a \neq 0$.	3	2												